Customer Shopping Behavior Analysis

Python | PostgreSQL | Power BI

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across four product categories (Clothing, Accessories, Footwear and Outerwear). The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

The work followed a three-stage workflow: the data was cleaned and prepared in **Python**, loaded into **PostgreSQL** for structured business analysis with ten SQL queries, and finally presented in an interactive **Power BI** dashboard.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18
- **Customer demographics:** Age, Gender, Location, Subscription Status
- **Purchase details:** Item Purchased, Category, Purchase Amount (USD), Season, Size, Color
- **Shopping behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method
- **Missing data:** 37 values in the Review Rating column (3,863 of 3,900 present)

Initial summary statistics showed purchase amounts ranging from $20 to $100 with a mean of about $59.76, customer ages from 18 to 70 (mean about 44), an average review rating of about 3.75, and an average of about 25 previous purchases per customer. Males accounted for 2,652 of the 3,900 records, and 2,223 records had no discount applied.

3. Exploratory Data Analysis using Python

Data preparation and cleaning were carried out in Python with the following steps:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.head()`, `df.info()` and `df.describe(include='all')` to check structure, data types and summary statistics.
- **Missing Data Handling:** Checked for null values with `df.isnull().sum()` and imputed the missing `Review Rating` values using the median rating of each product category (group-wise median imputation). A second null check confirmed no missing values remained.
- **Column Standardization:** Converted all column names to lowercase snake case (for example `Review Rating` became `review_rating`) and renamed `purchase_amount_(usd)` to `purchase_amount` for easier querying and documentation.

Feature Engineering:

- Created the **age_group** column by dividing customer ages into four quartile-based bins with `pd.qcut`, labelled Young Adult, Adult, Middle Aged and Senior. Based on the age quartiles (31, 44 and 57), these correspond to roughly 18-31, 32-44, 45-57 and 58-70 years.

- Created the **purchase_frequency_days** column by mapping the text frequency of purchases to a number of days between purchases.

Frequency of Purchases	purchase_frequency_days
Weekly	7
Fortnightly / Bi-Weekly	14
Monthly	30
Quarterly / Every 3 Months	90
Annually	365

- **Data Consistency Check:** Verified whether `discount_applied` and `promo_code_used` were redundant. The two columns were identical in every row, so `promo_code_used` was dropped.
- **Database Integration:** Connected the Python script to PostgreSQL using SQLAlchemy and psycopg2, and loaded the cleaned DataFrame into a table named `customer` in the `customer_behavior` database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

Structured analysis was performed in PostgreSQL to answer ten key business questions. The full queries are listed in the Appendix.

1. Revenue by Gender

Compared total revenue generated by male versus female customers.

Gender	Revenue (USD)
Female	75,191
Male	157,890

Insight: Male customers generated about 67.7% of revenue ($157,890 of $233,081). This closely mirrors their share of customers (2,652 of 3,900, about 68%). Average spend per customer is almost the same, roughly $59.54 for males and $60.25 for females, so the gap comes from the number of male customers and not from higher spending.

2. High-Spending Discount Users

Identified customers who used a discount but still spent at or above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
...	...

Insight: The query returned 839 customers (first rows shown). These customers bought with a discount applied yet still spent at or above the dataset average of about $59.76, which suggests some discounts may be given to buyers who would have purchased anyway.

3. Top 5 Products by Rating

Found the products with the highest average review ratings.

Item Purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Insight: Gloves lead with 3.86. The top five sit in a narrow band (3.78 to 3.86) only slightly above the overall average of 3.75, so product satisfaction is fairly uniform across the catalog.

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

Shipping Type	Average Purchase (USD)
Standard	58.46
Express	60.48

Insight: Express shipping customers spend about $2.02 more per purchase (around 3.5% higher) than Standard shipping customers.

5. Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

Subscription Status	Total Customers	Avg Spend (USD)	Total Revenue (USD)
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

Insight: Subscribers are 27% of customers and contribute about 26.9% of revenue. Average spend is essentially the same for both groups ($59.49 vs $59.87), so subscription does not currently translate into higher spend per purchase.

6. Discount-Dependent Products

Identified the five products with the highest percentage of purchases made with a discount.

Item Purchased	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Insight: About half of all purchases of these five items use a discount. Hats stand out as both the most discount-dependent product and one of the top-rated items (3.80).

7. Customer Segmentation

Classified customers into New, Returning and Loyal segments based on their number of previous purchases: New = 1 previous purchase, Returning = 2 to 10, Loyal = 11 or more.

Customer Segment	Number of Customers	Share
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%

Insight: Nearly 80% of customers are Loyal, while only 83 are New. The customer base is highly established, and growth will depend on acquiring new customers as much as on retaining existing ones.

8. Top 3 Products per Category

Listed the most purchased products within each category.

Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Insight: Order counts are very close within each category (for example Blouse and Pants tie at 171), so no single product dominates. Because ranks are assigned with ROW_NUMBER, tied products are ordered arbitrarily.

9. Repeat Buyers & Subscriptions

Checked whether customers with more than 5 previous purchases are more likely to subscribe.

Subscription Status	Repeat Buyers
No	2,518
Yes	958

Insight: Among the 1,053 subscribers, 958 (about 91%) are repeat buyers, compared with 2,518 of 2,847 non-subscribers (about 88%). Subscribers are only slightly more likely to be repeat buyers, so the link between repeat purchasing and subscription is weak.

10. Revenue by Age Group

Calculated the total revenue contribution of each age group.

Age Group	Total Revenue (USD)	Share
Young Adult	62,143	26.7%
Middle Aged	59,197	25.4%
Adult	55,978	24.0%
Senior	55,763	23.9%

Insight: Young Adults contribute the most revenue, and Young Adults plus Middle Aged customers together account for about 52% of the total. Revenue is spread fairly evenly across all four groups.

5. Dashboard in Power BI

Finally, an interactive **Power BI** dashboard was built to present the insights visually.



Figure 1: Customer Behavior Dashboard (Power BI)

Dashboard components

- **Filters (slicers):** Subscription Status, Gender, Category and Shipping Type (2-Day Shipping, Express, Free Shipping, Next Day Air, Standard, Store Pickup), which let users explore any segment of the data.
- **KPI cards:** 3.9K customers, $59.76 average purchase amount, and 3.75 average review rating.
- **% of Customers by Subscription Status:** 27% subscribed and 73% not subscribed.
- **Revenue by Category and Sales by Category:** Clothing is highest on both measures, followed by Accessories, Footwear and Outerwear.
- **Revenue by Age Group and Sales by Age Group:** Young Adults lead on both, followed by Middle Aged customers; Adult and Senior groups are close behind.

6. Business Recommendations

- **Boost Subscriptions:** Only 27% of customers subscribe and their average spend is no higher than non-subscribers. Promote exclusive benefits (early access, free shipping, members-only pricing) to make subscription more attractive and to lift spend per purchase.
- **Customer Loyalty Programs:** Reward repeat buyers to move the 701 Returning customers into the Loyal segment. With only 83 New customers, acquisition campaigns are also needed to refill the pipeline.
- **Review Discount Policy:** Products such as Hats, Sneakers, Coats, Sweaters and Pants are discounted in 47% to 50% of purchases, and 839 discount users still spent at or above average. Balance sales boosts with margin control by targeting discounts more selectively.
- **Product Positioning:** Highlight top-rated products (Gloves, Sandals, Boots, Hat, Skirt) and best-selling products (such as Jewelry, Blouse, Pants, Sandals and Jacket) in campaigns.

- **Targeted Marketing:** Focus efforts on the high-revenue Young Adult and Middle Aged groups and on Express-shipping users, who spend more per purchase. Female customers spend slightly more per customer yet make up only about a third of revenue, which points to room to grow that segment.

Appendix: SQL Queries

All queries were run in PostgreSQL against the `customer` table.

Q1. Revenue by gender

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

Q2. High-spending discount users

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

Q3. Top 5 products by average rating

```
SELECT item_purchased,
       ROUND(AVG(review_rating::numeric), 2) AS "Average Product Rating"
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

Q4. Standard vs. Express shipping

```
SELECT shipping_type, ROUND(AVG(purchase_amount), 2)
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Q5. Subscribers vs. non-subscribers

```
SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount), 2) AS avg_spend,
       ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

Q6. Discount-dependent products

```
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)
             / COUNT(*), 2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

Q7. Customer segmentation

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE
      WHEN previous_purchases = 1 THEN 'New'
      WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
      ELSE 'Loyal'
    END AS customer_segment
  FROM customer
)
SELECT customer_segment, COUNT(*) AS "Number of Customers"
FROM customer_type
GROUP BY customer_segment;
```

Q8. Top 3 products per category

```
WITH item_counts AS (
  SELECT category, item_purchased,
         COUNT(customer_id) AS total_orders,
         ROW_NUMBER() OVER (PARTITION BY category
                            ORDER BY COUNT(customer_id) DESC) AS item_rank
  FROM customer
  GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;
```

Q9. Repeat buyers and subscriptions

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Q10. Revenue by age group

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```